

K-Bound, Plainly

A reader’s companion to the short paper
“K-Bound: Knowing When Not to Adapt”

Companion narrative to the manuscript of Pratik Niroula

July 2026

This document is a plain-language retelling of the 23-page conference manuscript. It changes no claims, adds no results, and softens no caveats — it only translates. Each section notes where the real content lives in the paper, so you can read the two side by side. Where the paper is careful, this companion is careful in the same places.

The whole story in one page

Imagine a machine-learning model shipped into the world — a classifier reading medical slides, or a camera trap identifying animals. The world it meets is never quite the world it was trained on. A popular fix is *test-time adaptation* (TTA): let the model quietly update itself on the unlabeled data it sees, so it can track the drift.

Here is the uncomfortable part. The same self-update that rescues the model under one kind of shift can quietly wreck it under another — and because nobody has labels at deployment, *the system cannot see which of the two is happening*. Adaptation might be medicine or poison, and the patient can’t tell.

Nearly all research asks: *how* should the model adapt? This paper asks the question that comes first: **should it adapt at all?** It treats that as a decision with three possible answers — ADAPT (the update provably helps), FREEZE (it provably hurts, keep the old model), and ABSTAIN (the unlabeled evidence cannot tell, so don’t commit). The paper then does three things:

One, it draws the exact boundary of the knowable. There is a certain observable signal strength (M , the “margin”) and a declared amount of possible miscalibration (β , the “drift budget”). The paper proves the direction of the benefit is knowable *exactly when* the signal is stronger than the worst-case drift: $|M| > \beta$. Inside that band, two worlds can look *identical* in every label-free measurement and yet have *opposite* truths — so any rule that always commits must be wrong somewhere, and abstaining is not cowardice but mathematics.

Two, it builds a practical certificate. You never get to see M or the drift at deployment. So the method, KGA, does something humbler: it *predicts* the benefit from label-free signals,

wraps a pre-calibrated uncertainty interval around the prediction, and only commits when the entire interval is on one side of zero. That single discipline gives a provable cap on the rate of harmful commitments.

Three, it is honest about what the experiments show. On every natural distribution shift tested, the certificate is *no-harm*: it matches whichever trivial policy (always-adapt or always-freeze) happened to be right, while avoiding the wrong one — like insurance that never made things worse. Beating *both* trivial policies only happens where helpful and harmful conditions are mixed together and detectably so — on controlled stress grids and one constructed mixture — and the paper says so plainly rather than dressing those up as real-world victories.

That is the whole paper: a boundary, a certificate, and an honest map of where each holds.

1 A model alone in the wild

[paper: §I, Fig. 1]

Two characters carry the story. The **frozen model** f_0 is the trained network as it shipped — reliable, known, possibly stale. The **candidate** f_a is what the adapter (Tent, EATA, or SAR — three standard TTA methods) proposes after nudging the model on a batch of unlabeled deployment data.

The referee between them is a single number: the **benefit** $\Delta = R_T(f_0) - R_T(f_a)$ — old model’s risk minus new model’s risk on the deployment distribution. Positive Δ : the update helps, commit it. Negative: it hurts, roll it back. Zero: the two are equivalent. Everything in the paper is about one question: *can you learn the sign of Δ without labels?*

One detail makes the problem tractable and is worth absorbing early: the two models only differ where they *disagree*. On every input where f_0 and f_a say the same thing, committing or not makes no difference. So the whole contest lives on the **disagreement region** D — and there, for binary decisions, exactly one of the two models is right on each point. The benefit question becomes: *inside the disagreement region, which model is right more often?*

2 Three doors, and why the third one matters

[paper: §III.B]

The controller returns one of three actions. ADAPT commits the candidate. FREEZE rejects it and keeps the frozen model. ABSTAIN also keeps the frozen model — but it means something different: “the evidence could not certify either direction.” An abstain can page a human, request more data, or simply wait.

A crucial clarification the paper repeats, because people mishear it: abstention applies to the *update decision*, never to prediction. No input goes unanswered; the frozen model always serves.

What is withheld is only the *commitment* to a new model state.

Why dignify “do nothing” as a first-class action? Because the theory will show there is a regime where doing anything else is provably unsound. A two-action system *must* gamble there; a three-action system gets to say “I don’t know” — and saying it is the correct answer.

3 The rules of evidence

[paper: §III.C, Table II]

“Label-free” is a precise contract here, not a vibe. Source-domain labels and a held-out *development* set may be used — that is where the benefit predictor is trained and its uncertainty calibrated. But no label from the deployment test data may touch anything: not the adapter’s settings, not the evidence features, not the estimator, not the radius, not the decision threshold. Test labels are opened exactly once, *after* all decisions are frozen, to audit how the decisions went.

The paper’s Table II is the data-access ledger stage by stage. This discipline is why the false-adapt numbers later can be trusted: nothing downstream of the decision ever saw the answers.

4 The heart of the theory: a tug-of-war

[paper: §IV, Theorems 1–2, Fig. 2]

Strip away the formalism and the central theorem is a tug-of-war.

On one side pulls the **margin** M : how strongly the observable, label-free evidence points toward “the candidate is better” or “the frozen model is better” on the disagreement region. On the other side pulls **calibration drift** γ : the unobservable amount by which deployment reality can differ from what your calibration taught you — confidence that no longer means what it meant at home. You never see γ ; you only declare a *budget* β , a promise that $|\gamma| \leq \beta$ (“things won’t be more miscalibrated than this”).

The sign of the true benefit turns out to satisfy $\text{sign } \Delta = \text{sign}(M + \gamma)$. Read the tug-of-war off that equation:

- If $|M| > \beta$: the evidence pulls harder than the worst the hidden drift can pull back. The direction is settled *no matter what* γ does. A strict adapt or freeze commitment is uniformly sound. **This is the knowable regime.**
- If $|M| \leq \beta$: the hidden drift is strong enough to flip the outcome either way. And this is not just uncertainty — the paper constructs *twin worlds*: two deployment distributions that produce *exactly the same* label-free evidence, both allowed by the declared budget,

with *opposite* benefit signs. Any rule that commits in that band is wrong in one twin. **This is the unknowable regime, and abstention is forced.**

The boundary case $|M| = \beta$ gets its own careful treatment (zero-benefit ambiguity still spoils strict commitment), which is why the paper says “closed band.” Together this is Theorem 2, the *exact* frontier: knowable if and only if $|M| > \beta$. Not “roughly when,” but *iff* — the boundary is drawn, not sketched.

One more honest note the paper insists on: this frontier is a *population characterization*, a statement about what is in principle decidable. It is not a formula you can evaluate at deployment, because M is a population quantity and γ is invisible. Bridging that gap is the method’s job.

5 From theory to practice: KGA

[paper: §V, Algorithm 1, Fig. 3]

Since you cannot test $|M| > \beta$ directly, KGA (Knowability-Guided Adaptation) replaces the population tug-of-war with a finite-sample one you *can* run:

1. Let the adapter propose its candidate on the unlabeled batch (on a shadow copy — the live model is untouched).
2. Compute a vector Z of label-free signals: entropies, confidences, how much the two models disagree, how far the output distribution drifted, how big the parameter update was. Eleven features on the stress grids; sixteen on the natural shifts.
3. Feed Z to a small trained regressor that outputs $\hat{\Delta}$ — a *prediction* of the benefit.
4. Around that prediction, place a pre-calibrated radius ε , forming the interval $\hat{\Delta} \pm \varepsilon$.
5. Decide: interval entirely above zero $\Rightarrow$ ADAPT; entirely below $\Rightarrow$ FREEZE; straddles zero $\Rightarrow$ ABSTAIN (roll back, keep f_0).

The guarantee is disarmingly simple. Calibrate ε so that, with probability at least $1 - \alpha$, the true benefit lies inside the interval. On that “good event,” if the whole interval is positive then the truth is positive — so a harmful commit can only happen when the interval misses, which happens at most an α fraction of the time. That is Theorem 3: the *marginal* false-adapt rate is capped at α . (Marginal, not conditional — the paper is explicit that it does *not* promise “of the times we adapted, at most α were wrong.” That stronger conditional statement is not claimed.)

Three distinctions the paper guards jealously, because each is a place a careless reader would overclaim:

ε **is not** β . The radius is an empirical creature — a quantile of prediction errors measured on calibration data. The budget is a declared property of the deployment class. They are different objects; one never estimates the other.

The calibration is out-of-fold, always. For every calibration cell, the predictor scoring it was trained on the *other* cells; on natural shifts, the fit happens on development domains and the held-out domains are scored exactly once. No residual is measured by a model that saw its own answer. (An earlier version of the project got this wrong in places, found it in an audit, and corrected every affected result — see Section 7.)

Refit per adapter, per dataset. A benefit model calibrated for Tent does not transfer to SAR — the paper’s own ablation shows transferring it can push false-adapt as high as 25%. So the deployed method simply forbids that: each adapter and dataset gets its own fit. The safety claim is honest *because* its preconditions are enforced.

Operationally, the whole controller adds about 0.2ms per decision and one model copy in memory for rollback. The expensive part of TTA — computing the candidate — happens anyway; the certificate is a cheap gate bolted onto it.

6 What actually happened in the experiments

[paper: §VI–VII, Tables IV–XV]

The paper organizes results by *regime*, not by dataset, because the theory predicts different correct behavior in each. Keep the tug-of-war in mind and every row makes sense.

6.1 The synthetic dress rehearsal

[paper: §VII.A, Fig. 5]

First, a world where the authors control the truth: synthetic data where M , β , and the hidden drift are all known. Three predictions get checked. The benefit predictions cover at the promised rate (90% nominal, 90.0% realized). The commit rate slams from 6.5% inside the band to 96.7% outside it — the transition happening *at* the theoretical boundary, with the committed direction right 99.3% of the time. And the false-adapt rate stays under budget at every tested β . The frontier is not imposed anywhere in the code; it *emerges*. That is what validating a theory looks like.

6.2 Stress grids: the mixed regime, where routing wins

[paper: §VII.B–VII.E, Tables IV–VI]

The CIFAR-10-C *stress* grid deliberately mixes conditions where adaptation helps with conditions where it collapses (432 conditions per adapter, five seeds; about a third of cells are

harmful). This is exactly the regime where a router should shine, and it does: KGA adapts on the helpful cells, freezes the harmful ones, abstains on the marginal ones — with *zero* observed false-adapts in 2160 cell-decisions — and its regret beats *both* trivial policies with confidence intervals clear of zero.

Two comparisons sharpen this. Against *simple* label-free gates (trust confidence, trust entropy, trust a drift threshold, trust an accuracy estimate): the confidence and entropy gates false-adapt on roughly three quarters of the harmful cells — they trust exactly the signal that collapse inflates. Against *KGA minus its radius* (just “adapt if the predicted benefit is positive”): regret gets slightly better, but false-adapt jumps from 0 to about 5% — with no guarantee behind it. That single row is the thesis in miniature: **the estimator buys accuracy; the radius buys safety**.

The harder test is the **pre-registered head-to-head** against decision baselines in the style of POEM and AETTA — the nearest published competitors. Benchmark, metrics, and the win criterion were locked *before* any number was computed. KGA wins: lower regret than both, intervals clear of zero, while the ported baselines false-adapt on about a quarter of the stream. The paper is scrupulous about the asterisk: these are *protocol-matched ports* of the competitors’ ideas, not their official code — faithful to the setting, documented in a dedicated audit table, with official-repo runs named as remaining work. The claim is exactly as big as the evidence, and no bigger.

On ImageNet-C with SAR as the candidate (five seeds), the same pattern at scale: SAR harms on a detectable slice of conditions, and KGA routes around the damage, beating both fixed policies with zero false-adapts. With EATA, adaptation is nearly always helpful there — so KGA correctly just ties always-adapt. Same certificate, different regime, different (correct) behavior.

6.3 Natural shifts: the one-sided regime, where no-harm is the win

[paper: §VII.F, Tables VIII–X, Fig. 7]

Now the real world: hospital shift (Camelyon17), wildlife cameras (iWildCam), laboratory batches (RxRx1), domain style (Office-Home). Here a fact of nature takes over: a single natural benchmark is usually *one-sided*. Adaptation helps almost everywhere on it, or hurts almost everywhere. One of the two trivial policies is therefore already near-perfect — and *nothing* can meaningfully beat a policy that is already optimal.

So what should a *valid* certificate do there? Match the good policy, dodge the bad one, and keep its promise about false adaptation. That is precisely what happens, on every natural shift tested: on Camelyon17 adaptation is safe and KGA behaves like always-adapt; on RxRx1 adaptation is catastrophic (regret 0.25 if you always adapt) and KGA freezes everything; on iWildCam and Office-Home it crushes blind adaptation and ties the freeze policy. False adaptation: at or near zero throughout.

The paper names this outcome *no-harm* and calls it — deliberately, in the abstract, in the introduction, and again in the results — the **primary** finding, with the stress-grid wins as *secondary mechanism confirmation*. The framing matters: an operator deploying TTA wants exactly this contract (“you’ll get the good outcome and never the disaster”), and a certificate that manufactured “wins” on one-sided data would, by that very fact, be broken.

6.4 The boundary cases, kept in the picture

[paper: §VII.G–VII.H]

Two tracks are retained precisely because they refused to flatter the method. On **ImageNet-R**, evidence is weak; no candidate clears the bar on any consistent basis, and the method’s abstention there is presented as a *diagnostic*, with the paper explicitly declining to say *why* (weak evidence, poor estimation, and calibration failure are all compatible with the observation). On **CIFAR-10.1**, cross-seed calibration transfer fails the declared bar ($FA_u = 0.167$), and the paper simply reports the failure. PACS runs three seeds and lands as a null. A constructed three-source mixture shows the routing mechanism working across heterogeneous conditions — labeled loudly as a *constructed* aggregate, not a transfer result.

7 The honesty layer is part of the design

[paper: §VI.F, §IX, Tables XVI–XVII]

Most papers have results; this one also has a *chain of custody*. Every headline number carries a tier — **locked** (traces to raw, frozen artifacts), **reconciled** (corrected using raw records), **provisional**, or **diagnostic** — and only the first two support claims. Protocols were pre-registered before numbers existed. When an audit found that some early natural-shift “wins” had used an in-sample radius, the wins were withdrawn and reclassified as no-harm — the correction is described in the paper itself. When one CIFAR-10-C SAR aggregate would not reproduce from its raw files, the entire panel row was *withheld* rather than patched. A claim-to-support table maps each statement to its theorem or artifact. And a boxed paragraph states what is *not* claimed: no universal accuracy gains, no wins where always-adapt is already optimal.

There is also a machine-checked layer: core steps of the theory are formalized in Lean 4, with the paper carefully enumerating which clauses the kernel actually verified and which remain manuscript-proved — and explicitly declining to claim priority for it.

Read cynically, all this bookkeeping is overhead. Read correctly, it is the product: a paper about *knowing when not to trust* had better demonstrate the discipline it preaches, and this one does.

8 Limitations, in plain words

[paper: §VIII]

The paper’s own list, translated. The wins live where harm is *frequent, detectable, and costly*; on benign shifts you are buying insurance, and the premium is some abstention. The guarantee is exactly as strong as its three assumptions — interval coverage, calibration-to-deployment transfer (exchangeability), and risk-aligned evidence — and the paper tells you what breaks and what to do (widen, recalibrate, abstain) when each fails. The drift budget β must be *declared* from domain knowledge; it provably cannot be estimated from unlabeled data alone (label-free audits of it are vacuous — proved in the appendix), though a small labeled audit or a structural assumption can buy a computed bound. Abstention can come from four different causes (true unknowability, small samples, a weak estimator, failed calibration) and the paper refuses to let a null result claim the deepest cause. And the strongest baselines are ports, not official code — the head-to-head against official POEM/AETTA is named future work, not assumed.

9 A reader’s map

If you read the real paper next to this companion: §I and Fig. 1 give the decision at a glance; Table I is the regime map that organizes everything; §IV is the tug-of-war made rigorous (read Theorem 2’s statement, skim the constructions); §V is the method — Algorithm 1 plus the “population frontier versus empirical certificate” subsection, which is the paper’s most load-bearing honest distinction; §VII reads fastest regime-by-regime in the order this companion used; and the limitation and claim-accounting tables (§VI.F, §VIII–IX) are where the paper’s character lives.

Symbol	In plain words	Can you see it at deployment?
f_0, f_a	the frozen model; the adapter’s candidate	yes
Δ	how much committing would truly help (or hurt)	no — it is the thing being decided
D	the inputs where the two models disagree	yes
M	how strongly the evidence points one way	population quantity, via evidence
γ	the hidden calibration drift	never
β	declared worst case for $ \gamma $	you choose it, before deployment
Z	the label-free signal vector	yes
$\hat{\Delta}$	the predicted benefit	yes
ε	the calibrated uncertainty radius	fixed before deployment
α	the error rate you agree to tolerate	you choose it

Table 1: *

The cast of symbols, translated. The heart of the theory: knowable iff $|M| > \beta$. The heart of the method: act only when $\hat{\Delta} \pm \varepsilon$ clears zero.

The one-sentence version

K-Bound turns “should the model adapt?” from a gamble into a certified decision — proving exactly when the question is answerable, abstaining when it is not, and backing the whole thing with pre-registered evidence that is honest about being safety, not magic.